

PHYSICS 7B – Spring 2022
Final Exam, C. Bordel
Tuesday, May 10th, 8-11 am

• Name:

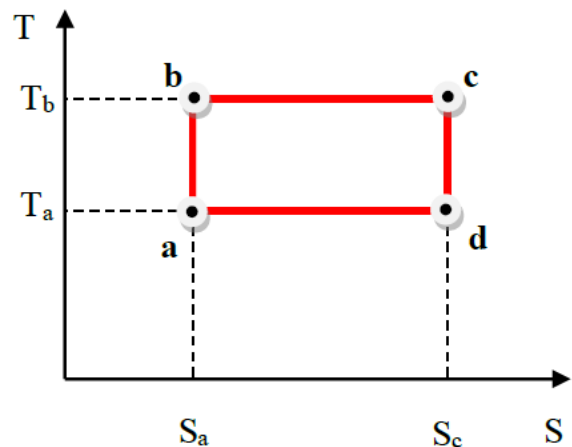
• Student ID #:

• Section #:

- **Rules:** *You may work on this exam from 8:00-11:00 am PT. This midterm is closed book and closed notes and you are **not** allowed to use any electronic devices. You are **not** allowed to communicate with any other students or discuss the content of the exam with anyone besides the Physics 7B teaching staff. Any violations of this policy will be considered a breach of academic integrity.*
- **Equation sheet:** *Only the equations given on the equation sheet can be used without derivation.*
- **Partial/Full Credit:** *We will give partial credit on this midterm, so if you are not sure how to do a problem, or if you do not have time to complete a problem, try to demonstrate your understanding of the physics involved by drawing a clear diagram and explaining (in physics terms) why you believe your answer to be incorrect, or how you would do the problem if you had time. Please be aware, however, that writing irrelevant or incorrect information will result in less partial credit. Remember that you need to show your work and justify your answers in order to get full credit.*
- **Honors Statement:** *It is expected that during this examination, as with any examination that they undertake at the university, students adhere to the usual standards of academic integrity at the University of California at Berkeley as outlined on by the [Center for Teaching and Learning](#). Therefore, by submitting your exam, you are affirming the following statement:*
“I swear on my honor that I have neither given nor received aid on this exam. In addition, I abided by all of the examination policies as outlined above.”

Problem 1 – 25 pts

While P-V diagrams are useful to describe the work done by the system in a thermodynamic cycle, temperature-entropy (T-S) diagrams are more useful to describe the heat absorbed by the system. In a T-S diagram, the temperature of a thermodynamic process is plotted as a function of the entropy of the process, as shown in the figure below. Fig.1 shows a thermodynamic cycle made up of four thermodynamic processes.



- a) Determine the nature (isobaric, isovolumetric, isothermal, or adiabatic) of the four processes plotted on the figure. What is the name of such a cycle?
- b) What does the area of the loop represent? What should be the direction of the cycle if it applies to a heat engine?

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- c) Determine the net mechanical work done by the gas in the cyclic process, using the T-S diagram above and the given variables.

- d) Calculate the efficiency of the cycle using the T-S diagram above, and express it in terms of the given variables.

Problem 2 - 25 points

A mass spectrometer, used to measure ion masses, is shown in cross-section view in Fig. 2. Ions of mass m and charge q are emitted with velocity $\vec{v}$ pointing to the right. As they pass through slit S_1 , they enter a region of crossed electric and magnetic fields. The two slits S_1 and S_2 are along the same horizontal line. The electric field $\vec{E}$ is created by the two horizontal charged plates, noted with (+) and (-). The magnetic field in this region is called $\vec{B}_1$ and points into the page, as shown by the crosses in Fig.2. The ions that pass through slit S_2 experience a magnetic field $\vec{B}_2$, also pointing into of the page.

Both magnetic fields are uniform. The ions are finally collected on a detector that sits on the plane (perpendicular to the page) containing S_2 . *You may assume that the gravitational force is negligible.*

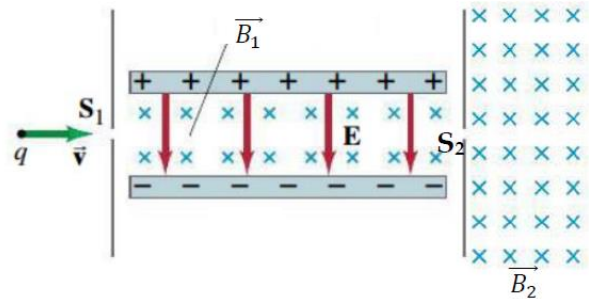


Figure 2

- a) Determine the speed v that allows the ions to go through both slits. Explain why the first chamber, between the two slits, is called a velocity selector.
- b) Explain the trajectory followed by the ions after the 2nd slit and determine where the ions are expected to hit the detector depending on the sign of their electric charge.

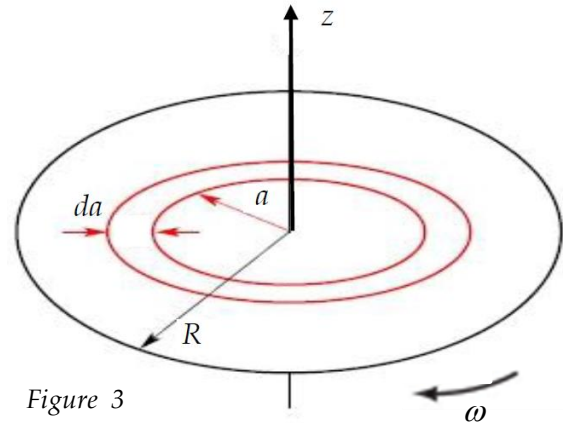
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- c) Determine the mass m of the ions in terms of the fields E , B_1 and B_2 , the ion charge q , and the distance d between S_2 and the detector.

- d) Is it possible to distinguish an ion of mass m and charge q from an ion of mass $2m$ and charge $2q$? Explain.

Problem 3 – 25 pts

A non-conducting circular disk of negligible thickness and radius R carries a non-uniform surface charge distribution $\sigma(a) = \sigma_0 \frac{a}{R}$, where σ_0 is a positive constant and a a variable measuring the radial distance from the center of the disk. The disk spins about its symmetry axis (z) in the clockwise direction viewed from the top, with angular speed ω , as shown in Fig.3.



- a) Determine the infinitesimal magnetic dipole moment of a thin circular ring of inner radius a and width da . *Hint: consider the infinitesimal current along the thin spinning ring.*

- b) Determine the magnetic dipole moment of the entire spinning disk.

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- c) Show that the magnitude of the magnetic field created by a circular loop of current (current I , radius a) on its symmetry axis (z -axis) is $\mathbf{B}(\mathbf{z}) = \frac{\mu_0 I a^2}{2(a^2 + z^2)^{3/2}}$.

- d) Determine the total magnetic field created on axis at distance $z \gg R$ as a function of the magnetic dipole moment.

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Problem 4 – 25 pts

A conducting loop carries current I in the counterclockwise direction. It is made of two pairs of identical linear segments- two short ones of length R and two long ones of length $2R$ - at 90 degrees from one another, and a curved section spanning a quarter of a circle, as shown in Fig.4. Point P is the center of the circular section.

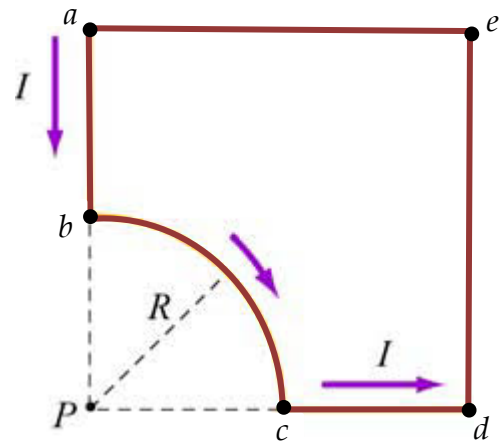


Figure 4

- a) Determine the magnetic field generated by the two *short* segments ab and cd at point P .
- b) Determine the magnetic field generated by the circular section (arc bc) at point P .

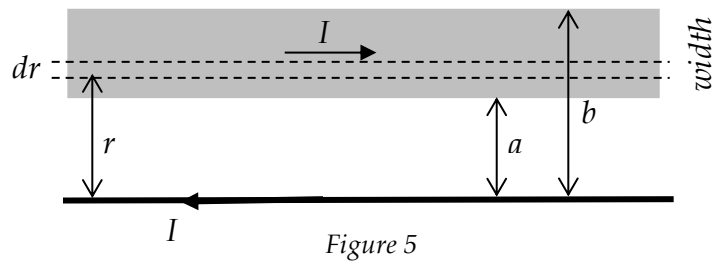
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c) Determine the magnetic field generated by the two *long* segments (*de* and *ea*) at point *P*.

d) Determine the magnitude and direction of the net magnetic field generated by the whole current-carrying loop at point *P*.

Problem 5 - 25 pts

A very long thin wire carries a current I to the left and a very long flat conducting strip is positioned parallel to the thin wire with its lower/upper edges at distance a and b from the wire, respectively. The strip carries the same current I as the wire, uniformly distributed over the width and pointing to the right (Fig.5).



- a) Determine the magnitude and direction of the infinitesimal magnetic force per unit length exerted by the wire on a very thin strip of width dr of the current-carrying strip.

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- c) Determine the magnitude and direction of the magnetic field created by the strip at the position of the wire. *Hint: Newton's third law might be helpful.*

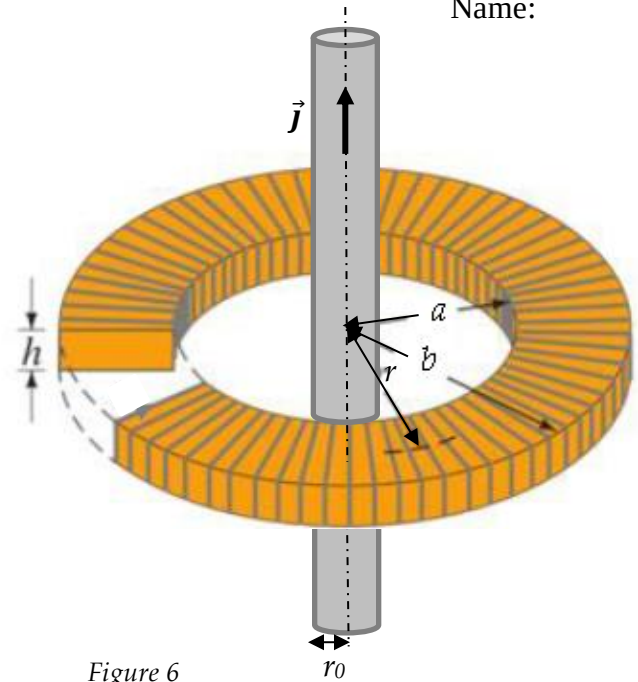
Let's assume that the wire and strip are clamped so that they cannot move, and let's consider that the strip is so narrow that the magnetic field created by the current-carrying wire is quasi-uniform over the width of the strip. The top of the strip is found to be at a lower electric potential than the bottom and the absolute value of the voltage measured across the width of the strip is V_H .

- d) Determine which type of charge carrier is responsible for the current in the strip and determine the drift velocity of the charge carriers.

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Problem 6 – 25 pts

A very long thick conducting wire of radius r_0 carries an upward uniform but time-dependent current density of magnitude $j = j_0 \exp(-t/t_0)$, where j_0 and t_0 are positive constants. A toroid is placed around the thick wire, with its center at the center of the wire and its symmetry axis along the thick wire. The toroid has N turns, inner radius a , outer radius b , and height h , as shown in Fig.6. The metallic wire that the toroidal coil is made of has resistivity ρ and diameter d . The coil is not connected to any power source but forms a closed loop. Note that the clear "slice" of the toroid is not indicative of a missing part but allows you to view the rectangular cross-section of the coil.



- a) Determine the magnetic field generated by the thick wire at any distance $r > r_0$.

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- b)** Determine the total flux of magnetic field through the toroid.

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c) Determine the absolute value and the direction of the induced current.

d) Determine the mutual inductance of the system. *Note that the thick wire is part a big current loop, whose other sides have a negligible contribution to the magnetic flux generated through the toroid.*

Thermal expansion

$$\Delta l = \alpha l_0 \Delta T$$

$$\Delta A = 2\alpha A_0 \Delta T$$

$$\Delta V = \beta V_0 \Delta T = 3\alpha V_0 \Delta T$$

Calorimetry

$$Q = mc\Delta T = nC\Delta T \text{ (no phase transition)}$$

$$Q = mL \text{ (for a phase transition)}$$

$$\sum_{i=1}^n Q_i = 0 \text{ (for a closed system)}$$

Heat conduction

$$H = \frac{dQ}{dt} = -kA \frac{dT}{dx}$$

Ideal gases

$$PV = Nk_B T = nRT$$

$$\frac{1}{2} m \overline{v^2} = \frac{3}{2} k_B T$$

$$v_{rms} = \sqrt{\overline{v^2}}$$

$$f_{Maxwell}(v) = 4\pi N \left(\frac{m}{2\pi kT} \right)^{\frac{3}{2}} v^2 e^{-\frac{mv^2}{2kT}}$$

$$\overline{g(v)} = \int_0^\infty g(v) \frac{f(v)}{N} dv$$

$$\Delta E_{int} = Q - W \text{ or } dE_{int} = dQ - PdV$$

$$W = \int PdV$$

$$E_{int} = \frac{d}{2} Nk_B T = nC_V T$$

$$C_V = \frac{d}{2} R$$

$$C_P - C_V = R = N_A k_B$$

$$\gamma = \frac{C_P}{C_V} = \frac{d+2}{d}$$

$$e = \frac{W_{net}}{Q_{in}} = 1 - \frac{Q_{out}}{Q_{in}}$$

$$e_{ideal} = 1 - \frac{T_L}{T_H}$$

$$\Delta S = \int \frac{dQ}{T} \text{ (For reversible processes)}$$

$$dQ = TdS$$

$$\Delta S_{syst} + \Delta S_{env}$$

$$> 0 \text{ (For irreversible processes)}$$

<i>Process</i>	Q	W	
Isobaric ($P=const$)	$nC_p\Delta T$	$P\Delta V$	
Isochoric ($V=const$)	$nC_v\Delta T$	0	
Isothermal ($PV=const$)	$nRT \ln\left(\frac{V_f}{V_0}\right)$	$nRT \ln\left(\frac{V_f}{V_0}\right)$	
Adiabatic ($PV^\gamma =$ $const$)	0	$\frac{(P_fV_f - P_0V_0)}{1 - \gamma}$	
Gas	C_v	C_p	γ
Monatomic	$\frac{3R}{2}$	$\frac{5R}{2}$	$\frac{5}{3}$
Diatomic	$\frac{5R}{2}$	$\frac{7R}{2}$	$\frac{7}{5}$
Polyatomic (non-linear)	$3R$	$4R$	$\frac{4}{3}$

Electrostatics

$$\vec{F} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \hat{r}$$

$$\vec{F} = Q\vec{E}$$

$$d\vec{E} = \frac{dQ}{4\pi\epsilon_0 r^2} \hat{r}$$

$$\lambda = \frac{dQ}{dl} ; \quad \sigma = \frac{dQ}{dA} ; \quad \rho = \frac{dQ}{dV}$$

$$\vec{p} = Q\vec{d}$$

$$\vec{\tau} = \vec{p} \times \vec{E}$$

$$U = -\vec{p} \cdot \vec{E}$$

$$\Phi_E = \int \vec{E} \cdot d\vec{A}$$

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

$$\Delta U = Q\Delta V$$

$$V(b) - V(a) = - \int_a^b \vec{E} \cdot d\vec{l}$$

$$dV = \frac{dQ}{4\pi\epsilon_0 r}$$

$$\vec{E} = -\vec{\nabla}V$$

$$\vec{F} = -\vec{\nabla}U$$

Capacitors

$$Q = CV$$

$$C_{eq} = C_1 + C_2 \text{ (In parallel)}$$

$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} \text{ (In series)}$$

$$C = KC_0$$

$$U = \frac{Q^2}{2C}$$

$$u = \frac{\epsilon}{2} |\vec{E}|^2$$

$$\epsilon = K\epsilon_0$$

Current

$$I = \frac{dQ}{dt}$$

$$I = \int \vec{j} \cdot d\vec{A}$$

$$\vec{j} = nQ\vec{v}_d = \frac{\vec{E}}{\rho}$$

Resistance

$$R = \int \rho \frac{dl}{A}$$

$$\rho(T) = \rho(T_0)(1 + \alpha(T - T_0))$$

$$R_{eq} = R_1 + R_2 \text{ (In series)}$$

$$\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} \text{ (In parallel)}$$

DC circuits

$$P = IV$$

$$V = IR \text{ (Ohm's law)}$$

$$\sum_{junc.} I = 0 \text{ (junction rule)}$$

$$\sum_{loop} V = 0 \text{ (loop rule)}$$

RC circuits

$$\text{Solution to } \frac{dQ}{dt} + \frac{Q}{RC} = \frac{\mathcal{E}}{R}$$

$$\text{is } Q(t) = C\mathcal{E}(1 - e^{-t/(RC)})$$

with $C\mathcal{E} = Q_{max}$ and $\tau = RC$

$$\text{Solution to } \frac{dQ}{dt} + \frac{Q}{RC} = 0$$

$$\text{is } Q(t) = C\mathcal{E}e^{-t/(RC)}$$

Magnetism

$$d\vec{F} = Id\vec{l} \times \vec{B}$$

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$\vec{\mu} = NI\vec{A}$$

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

$$U = -\vec{\mu} \cdot \vec{B}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{encl}$$

$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{I d\vec{l} \times \hat{r}}{r^2}$$

$$\Phi_B = \int \vec{B} \cdot d\vec{A}$$

Electromagnetic Induction

$$\mathcal{E} = \oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

$$\mathcal{E} = -L \frac{dI}{dt}$$

$$M = N_1 \frac{\Phi_1}{I_2} = N_2 \frac{\Phi_2}{I_1}$$

$$L = N \frac{\Phi}{I}$$

$$U = \frac{1}{2} LI^2$$

$$u = \frac{1}{2\mu_0} |\vec{B}|^2$$

LR circuits

$$\text{Solution to } \frac{dI}{dt} + \frac{RI}{L} = \frac{\mathcal{E}}{L}$$

$$\text{is } I(t) = \frac{\mathcal{E}}{R} (1 - e^{-Rt/L})$$

$$\text{with } \frac{\mathcal{E}}{R} = I_{\max} \text{ and } \tau = \frac{L}{R}$$

$$\text{Solution to } \frac{dI}{dt} + \frac{RI}{L} = 0$$

$$\text{is } I(t) = I_0 e^{-Rt/L}$$

Gradient and displacement

$$\vec{\nabla} f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{\partial f}{\partial z} \hat{z}$$

$$d\vec{l} = dr\hat{r} + r d\theta \hat{\theta} + r \sin(\theta) d\phi \hat{\phi}$$

(Cylindrical Coordinates)

$$\vec{\nabla} f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin(\theta)} \frac{\partial f}{\partial \phi} \hat{\phi}$$

$$d\vec{l} = dr\hat{r} + r d\theta \hat{\theta} + r \sin \theta d\phi \hat{\phi}$$

(Spherical Coordinates)

Areas and volumes

$$\text{Cylinder: } A = 2\pi RH ; V = \pi R^2 H$$

$$\text{Cylindrical shell: } dV = 2\pi R H dR$$

$$\text{Thin Circular Ribbon : } dA = 2\pi R dR$$

$$\text{Section of Thin Ribbon : } dA = R dR d\theta$$

$$\text{Sphere: } A = 4\pi R^2 ; V = \frac{4}{3}\pi R^3$$

$$\text{Spherical shell: } dV = 4\pi R^2 dR$$

Motion and Energy

$$\Delta K + \Delta U = W_{nc}$$

$$K = \frac{1}{2} m v^2$$

$$\vec{F}_{sp} = -k \vec{x} ; U_{sp} = \frac{1}{2} k x^2$$

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0 \text{ (simple harmonic motion)}$$

Integrals

$$\int_0^\infty x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

$$\int_0^\infty x^{2n} e^{-ax^2} dx = \frac{(2n)!}{n! 2^{2n+1}} \sqrt{\frac{\pi}{a^{2n+1}}}$$

$$\int_0^\infty x^{2n+1} e^{-ax^2} dx = \frac{n!}{2a^{n+1}}$$

$$\int (1+x^2)^{-\frac{1}{2}} dx = \arctan(x)$$

$$\int (1+x^2)^{-\frac{3}{2}} dx = \frac{x}{\sqrt{1+x^2}}$$

$$\int \frac{x}{1+x^2} dx = \frac{1}{2} \ln(1+x^2)$$

$$\int \frac{1}{\cos(x)} dx = \ln \left(\left| \tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right| \right)$$

$$\int \frac{1}{\sin(x)} dx = \ln \left(\left| \tan \left(\frac{x}{2} \right) \right| \right)$$

Derivatives

$$\frac{d((f(x))^n)}{dx} = n(f(x))^{n-1} \frac{df}{dx}$$

$$\frac{d(\tan x)}{dx} = \sec^2 x = \frac{1}{\cos^2 x}$$

$$\frac{d(\cotan x)}{dx} = -\csc^2 x = -\frac{1}{\sin^2 x}$$

Taylor Expansions near x=0

$$\sin(x) \approx x$$

$$\cos(x) \approx 1 - \frac{x^2}{2}$$

$$e^x \approx 1 + x + \frac{x^2}{2}$$

$$(1+x)^\alpha \approx 1 + \alpha x + \frac{(\alpha-1)\alpha}{2} x^2$$

$$\ln(1+x) \approx x - \frac{x^2}{2}$$

Trigonometry

$$\sin(2x) = 2 \sin(x) \cos(x)$$

$$\cos(2x) = 2 \cos^2(x) - 1$$

$$\sin(a+b) = \sin(a) \cos(b) + \cos(a) \sin(b)$$

$$\cos(a+b) = \cos(a) \cos(b) - \sin(a) \sin(b)$$

$$1 + \cot^2(x) = \csc^2(x) ; 1 + \tan^2(x) = \sec^2(x)$$

$$c^2 = a^2 + b^2 - 2ab \cos(\theta) \text{ (law of cosine)}$$

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